

Yuxiang Qiu

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Last updated in 2026-09-14

Education

University of Pennsylvania

2026 – Now

Ph.D. Computer and Information Science; Advisor: Sebastian Angel

University College London

09/2021 – 06/2025

MEng Computer Science; First Class (rank: 1/150)

Georgia Institute of Technology

08/2023 – 05/2024

BS Computer Science (Exchange); GPA 4.0/4.0

Publications

Zippel: A Language, Compiler, Runtime, and Analyzer for ZK Proof Systems

Sebastian Angel, Sydney-Shea Cohen, Eleftherios Ioannidis, Pratyush Mishra, Alex Ozdemir, Yuxiang Qiu, Alireza Shirzad
IEEE Symposium on Security and Privacy (S&P), 2027

A Toolbox for Modelling Engagement with Educational Videos

Yuxiang Qiu, Karim Djemili, Denis Elezi, Aaneel Shalman Srazali, Mar'ia P'erez-Ortiz, Emine Yilmaz, John Shawe-Taylor
and Sahan Bulathwela

Proceedings of the AAAI Conference on Artificial Intelligence, 2024

TrueLearn: A Python Library for Personalised Informational Recommendations with (Implicit) Feedback

Yuxiang Qiu, Karim Djemili, Denis Elezi, Aaneel Shalman, María Pérez Ortiz and Sahan Bulathwela

6th Workshop on Online Recommender Systems and User Modeling, ACM RecSys 2023

Research Experience

Zippel: A Language, Compiler, Runtime, and Analyzer for ZK Proof Systems

04/2026 – Now

University of Pennsylvania

Philadelphia, PA

- Proposed **Zippel**, a domain-specific language for implementing and formally analyzing ZK proof systems.
- Designed static-analysis pipelines over graph-based IR, converting to polynomial ideals for Gröbner-basis analysis.
- Developed special-soundness analysis for multi-round interactive proof systems, improving completeness analysis.

TensorZKP: Efficient Zero-Knowledge Proof Acceleration using Tensor Cores

09/2025 – 12/2025

National University of Singapore

Singapore

- Developed **TensorZKP**, mapping sumcheck and linear-time encoding onto tensor-core matrix multiplications.
- Integrated fixed-constant Barrett reduction (Shoup-style) into GEMM pipelines; implemented high-performance field arithmetic kernels for arbitrary primes with optimizations for Mersenne and pseudo-Mersenne structures.
- Engineered warp-specialized GEMM kernels for linear-time encoding, load-balanced via inverted expander graphs.

Mim: Trustless Efficient Light Clients

10/2024 – 04/2025

University College London

London, UK

- Designed **Mim**, a light-client protocol with constant/log-size committee-rotation proofs via folding SNARKs.
- Introduced a Merkle accumulator to reduce bootstrapping overhead and amortize proof costs over long histories.
- Implemented a full Rust system (>8,000 LOC) using arkworks and sonobe; evaluated on a Sui-like blockchain, enabling sub-12-hour single-day proofs and sub-7-day one-year history proofs with 1-3s verification time.

TaskEval: Using LLMs to Evaluate Natural Text Artifacts (Patch Explanations)

06/2024 – 09/2024

UCL Software Optimisation, Learning and Analytics Research Lab

London, UK

- Proposed **TaskEval**, a framework scoring patch explanations via downstream LLM performance vs. LLM-as-a-judge.
- Constructed an evaluation infrastructure integrating SWE-bench, with fault localization and differential testing.
- Evaluated agreement and ranking quality via Kendall's τ and Spearman correlation across text perturbations.

Work Experience

Software Development Engineer Intern

06/2023 – 08/2023

Amazon

London, UK

- Researched cross-platform portability of Java apps running on Windows, resulting in a ~10-page research report.
- Developed a Java application and library that performs **incompatibility detection at the bytecode** level (checking for 7 different types of cross-platform issues) with **~80% accuracy and 90%+ recall** on real-world JARs.
- Optimized libraries by profiling hot spots and bringing parallelism to CPU-bound tasks, resulting in a **3x speedup**.

Teaching Assistant

- UCL
- London, UK
- 2024-2025: COMP0002 Principles of Programming, COMP0004 Object-Oriented Programming
 - 2022-2023: Developed an open-source course, tutored 12 students in 6 languages, scripting, networking, and web dev.

Honors and Awards

UCL Computer Science Best Overall Performance - Year 4	2025
UCL Computer Science Studentship for Research	2024
UCL Faculty Undergraduate Scholarships for Excellence (1 student per faculty, 1 out of 1000+ students)	2022

Open Source Contributions

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- **AI:** pytorch/torcheval ([#195](#)), princeton-nlp/SWE-bench ([#186](#), [#189](#), [#212](#))
 - **Crypto:** arkworks-rs/r1cs-std ([#157](#), [#161](#), [#171](#))
 - **PL:** rust-lang/rust-clippy ([#11865](#), [#12084](#), [#12094](#)), typst/biblatex ([#34](#))